



GRADE 7

STEM OLYMPIAD

National Level Examination (NLE) 2026

SUBJECT CODE			
---------------------	--	--	--

Total Questions: 50

Total Marks: 50

Time: 75 Minutes

DO NOT OPEN THIS BOOKLET UNTIL INSTRUCTED TO DO SO

- **Carefully Check Your Name, Class, Roll Number, And Other Required Details On The OMR Answer Sheet.**
- **Do Not Forget To Sign The OMR Sheet. Also Write Your Roll No. On The Question Paper Booklet.**
- **Use Only Black/Blue Ball Pen And HB Pencil To Fill The OMR Answer Sheet.**
- **All Questions Are Compulsory And Carry One Mark Each.**
- **Section A: Mathematics & Quantitative Reasoning (15 Questions); Section B: Science & Scientific Reasoning (15 Questions); Section C: Engineering, Technology & Data (10 Questions); Section D: STEM Olympiad Challenge (10 Questions).**
- **Only OMR Answer Sheets Will Be Evaluated.**
- **No Negative Marking For Incorrect Answers.**
- **You May Use The Space Provided On The Last Page For Rough Work (If Needed).**
- **Digital Devices Like Calculators, Smart Watches, Or Mobile Phones Are Not Allowed.**
- **After The Exam, Submit Both The OMR Sheet And The Question Paper To The Invigilator.**

SECTION A – MATHEMATICS & QUANTITATIVE REASONING

Q1. What is $-\frac{3}{4} + \frac{5}{8}$?

- A. $-\frac{1}{8}$
- B. $\frac{1}{8}$
- C. $\frac{3}{8}$
- D. $\frac{7}{8}$

Q2. Solve: $3x + 7 = 4x - 5$.

- A. $x=5$
- B. $x=6$
- C. $x=7$
- D. $x=8$

Q3. What is 15% of 360?

- A. 45
- B. 54
- C. 60
- D. 72

Q4. Simplify: $3a + 2a - a$.

- A. $2a$
- B. $3a$
- C. $4a$
- D. $5a$

Q5. What is $2^5 \times 2^3$?

- A. 2^8
- B. 4^8
- C. 2^{15}
- D. 4^5

Q6. A circle has radius 7 cm. Using $\pi = \frac{22}{7}$, its circumference is:

- A. 22 cm
- B. 44 cm
- C. 77 cm
- D. 154 cm

Q7. A rectangle is 15 cm by 8 cm. If each side is doubled, its area becomes:

- A. 240 cm^2
- B. 360 cm^2
- C. 480 cm^2
- D. 960 cm^2

Q8. The mean of 12, 16, 18, 22 and 32 is:

- A. 18
- B. 20
- C. 22
- D. 24

Q9. Which rational number lies between $\frac{1}{2}$ and $\frac{3}{4}$?

- A. $\frac{2}{5}$
- B. $\frac{5}{8}$
- C. $\frac{4}{5}$
- D. $\frac{7}{8}$

Q10. A product is first increased in price by 20% and then discounted by 20%. Compared with its original price, the final price.

- A. The same
- B. 4% higher
- C. 4% lower
- D. 8% lower

Q11. Pipe A fills a tank in 6 hours, while pipe B fills it in 12 hours. If both are opened together, how long will they take to fill the tank?

- A. 3 hours
- B. 5 hours
- C. 4 hours
- D. 6 hours

Q12. A 4×4 grid contains numbers from 1 to 16, each used exactly once. The sum of every row and every column must be equal. What must be the common sum?

- A. 30
- B. 32
- C. 34
- D. 36

Q13. An exterior angle of a triangle equals:

- A. the sum of the two opposite interior angles
- B. one interior angle only
- C. 180° minus all angles
- D. half the largest angle

Q14. The probability of getting an even number on a fair die is:

- A. $\frac{1}{6}$
- B. $\frac{1}{3}$
- C. $\frac{1}{2}$
- D. $\frac{2}{3}$

Q15. A car travels 150 km in 2.5 h. Its average speed is:

- A. 50 km/h
- B. 60 km/h
- C. 65 km/h
- D. 75 km/h

SECTION B – SCIENCE & SCIENTIFIC REASONING

Q16. Two identical plants are kept under identical conditions except for carbon dioxide concentration.

Plants A receives normal atmospheric CO_2

Plants B receives significantly more CO_2

After several hours, plant B initially shows a higher rate of photosynthesis.

- A. Availability of light or water
- B. Presence of gravity
- C. Number of chromosomes
- D. Length of the plant's roots only

Q17. In humans, most nutrients are absorbed in the:

- A. stomach
- B. small intestine
- C. large intestine
- D. mouth

Q19. Rusting of iron is a:

- A. physical change
- B. chemical change
- C. change of state only
- D. reversible change

Q20. A metal spoon is left in a cup of hot tea. After some time, its handle becomes hot. Which process is mainly responsible.

- A. conduction
- B. convection
- C. radiation only
- D. reflection

Q18. A farmer's soil is found to be highly acidic. Which substance would generally be appropriate to help reduce the acidity?

- A. Vinegar
- B. Lime juice
- C. Soaked lime
- D. Carbonated water

Q21. Sea breeze occurs mainly because land and water:

- A. heat at different rates
- B. have the same temperature always
- C. do not absorb sunlight
- D. produce magnets

Q22. Which soil generally holds the most water?

- A. Sandy
- B. Clayey
- C. Gravel
- D. Rock only

Q23. A ring of bark is carefully removed from the stem of a tree without damaging the central woody region. After several weeks, swelling is observed above the ring. What is the best explanation?

- A. Water accumulates above the ring because xylem is blocked
- B. Food transported through phloem accumulates above the removed region.
- C. Photosynthesis stops completely below the rings.
- D. Roots begin producing extra food.

Q24. Respiration releases energy from:

- A. food
- B. soil
- C. sunlight directly in animals
- D. minerals only

Q25. A machine converts electrical energy into machinal energy with an efficiency of 75%. If it receives 800J of electrical energy, how much useful machinal energy does it produce?

- A. 200J
- B. 400J
- C. 600J
- D. 1,067J

Q26. An electromagnet becomes stronger when:

- A. more current flows through more coil turns
- B. the circuit is open
- C. iron core is removed and current reduced
- D. the wire is cut

Q27. The image formed by a plane mirror is:

- A. virtual and same size
- B. real and inverted
- C. always smaller
- D. always larger

Q28. Groundwater depletion is reduced by:

- A. rainwater harvesting
- B. paving all open land
- C. over-pumping wells
- D. cutting vegetation

Q29. A population of rabbits in a forest increases rapidly because predators become scarce. After some time, the rabbit population suddenly decreases. Which explanation is most likely?

- A. Increased competition for limited food and resource
- B. Rabbits stops requiring food after reproduction
- C. Predators are necessary for rabbits to breathe
- D. The forest produces unlimited food.

Q30. A gardener notices that plants produced from seeds show greater variation than plants produced through vegetative propagation. Why?

- A. Seeds are always larger than vegetative parts
- B. sexual reproduction involves genetic material from two parents, creating variation.
- C. Vegetative propagation changes the DNA of every new plant.
- D. Seeds contain no genetic information.

SECTION C – ENGINEERING, TECHNOLOGY & DATA

Q31. A truss bridge is strong because triangles:

- A. resist deformation
- B. can never break
- C. remove gravity
- D. generate electricity

Q32. A robot has 4 possible instructions:

F=move forward 2 steps

B=move backward 1 step

R= turn right

L= turn left

It starts facing north at position (0,0).

It performs F→R→F→L→F→B.

What is its final position?

- A. (2,3)
- B. (2,4)
- C. (1,3)
- D. (1,4)

Q33. A student measures a pendulum period. For a fair test of length, which variable should be changed?

- A. Length only
- B. Length and mass together
- C. Starting angle and length together
- D. Clock used each time

Q34. A machine has input energy 500 J and useful output 350 J. Its efficiency is:

- A. 35%
- B. 50%
- C. 70%
- D. 85%

Q35. A bicycle gear with 20 teeth drives one with 40 teeth. The driven gear rotates:

- A. half as fast
- B. twice as fast
- C. at the same speed
- D. four times as fast

Q36. Which graph best shows distance changing continuously with time?

- A. Line graph
- B. Pie chart
- C. Venn diagram
- D. Tally chart only

Q37. A sensor gives readings 2, 5, 10, 17, 26.... Which number should come next?

- A. 35
- B. 36
- C. 37
- D. 38

Q38. A robot moves 6 m East, 4 m North, 2 m West, 1 m South. Its displacement from start is:

- A. 4 m East and 3 m North
- B. 8 m East and 5 m North
- C. 4 m West and 3 m North
- D. 6 m East and 1 m North

Q39. A solar cooker is improved by adding a reflector because it:

- A. directs more solar radiation to the pot
- B. increases friction
- C. reduces sunlight
- D. creates fuel

Q40. A substance X reacts with oxygen and produces a new substance Y. Y cannot easily be converted back into X by simple physical methods. Which statement is most appropriate?

- A. X has undergone a chemical change
- B. X has only changed its state
- C. No new substance has formed
- D. The process must be melting.

SECTION D – STEM OLYMPIAD CHALLENGE

Q41. A 2 kg object is raised 5 m. Taking $g=10$ N/kg, its gain in gravitational potential energy is:

- A. 10 J
- B. 25 J
- C. 50 J
- D. 100 J

Q42. A plant loses 30 mL water by transpiration in 2 h. At the same rate, loss in 5 h is:

- A. 60 mL
- B. 75 mL
- C. 90 mL
- D. 150 mL

Q43. Which variable should be controlled when comparing how blade angle affects a wind turbine model?

- A. Wind speed
- B. Blade angle
- C. Measured output
- D. Research question

Q46. A mixture contains iron filings, sand and salt. What is the best sequence to separate all three?

- A. Use magnet, dissolve salt in water, filter sand, evaporate water
- B. Filter first, then magnet through water only
- C. Evaporate first without adding water
- D. Use only a sieve

Q47. A 1.5 kW heater runs for 2 h. Energy used is:

- A. 0.75 kWh
- B. 1.5 kWh
- C. 3 kWh
- D. 3000 kWh

Q48. A drone travels 120 m north then 50 m south. Its net northward displacement is:

- A. 50 m
- B. 70 m
- C. 120 m
- D. 170 m

Q44. An arch bridge carries load mainly by transferring force into:

- A. compression along the arch
- B. light
- C. electric charge
- D. sound

Q45. Which sequence is most scientific?

- A. Question - hypothesis - test - analyse - conclude
 - B. Conclude - guess - test - ignore data
 - C. Build - decorate - conclude
 - D. Ask - answer without testing
-

Q49. In a design test, Prototype A holds 8 kg, B holds 11 kg and C holds 10 kg before failure. Which currently has the greatest load capacity?

- A. A
- B. B
- C. C
- D. All equal

Q50. A fair coin is tossed twice. Probability of two heads is:

- A. $\frac{1}{2}$
 - B. $\frac{1}{3}$
 - C. $\frac{1}{4}$
 - D. $\frac{3}{4}$
-

Sample paper